ND-INNOVATION PORTFOLIO

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CONFIDENTIAL & PROPRIETARY

Patent-Pending Technologies | June 2026

EXECUTIVE SUMMARY

This portfolio presents five integrated, patent-pending production optimization technologies designed specifically for Niger Delta marginal and mature fields. The innovations span the entire production stack — from downhole mechanical hardware to surface gas-lift systems, AI-native artificial lift platforms, autonomous reservoir intelligence probes, and a closed-loop autonomous control architecture. Together, they form a unified platform that addresses the three critical barriers to marginal field viability: high intervention cost, unreliable artificial lift, and lack of real-time reservoir intelligence.

All five technologies have provisional patents filed (priority date: June 13, 2026). Full non-convention Nigerian patent applications are in progress, with PCT international strategy under development. The portfolio is currently under evaluation by the SPE Nigeria Council Patent Recognition Program.

5	3	Growth	12 mo
Patent-Pending Innovations	SPE Patent Applications	Seeking Growth Capital	Priority Window Remaining

01. NIGER DELTA SURVIVOR

Slimline Tubing Anchor Catcher for Marginal Fields

The Niger Delta Survivor is a slimline tubing anchor catcher engineered to address the specific mechanical failure modes prevalent in Niger Delta marginal fields: tubing movement, packer failure, and premature well shutdown. Unlike conventional anchor catchers designed for high-rate, high-pressure environments, the Survivor is optimized for low-rate, high-water-cut, corrosive conditions typical of mature West African assets.

The design incorporates a reduced outer diameter profile that enables deployment through worn tubing and tight restrictions, while maintaining the load-bearing capacity required for deep-set packer applications. Sensor-ready ports allow future integration with the ND-Amahor platform for real-time load and vibration monitoring.

Key Claims:

• Slimline profile optimized for marginal field tubing IDs

• Corrosion-resistant alloy selection for H_2S/CO_2 environments

• Modular slip design for variable casing weights

• Embedded sensor ports for ND-Amahor integration

• Indigenous fabrication specification

Patent Status: Provisional filed June 13, 2026. Non-convention Nigerian patent application submitted with multiple claims.

Commercialization: Targeting leading indigenous fabricators for exclusive fabrication licensing under industry-standard licensing terms.

02. ND-GLO

Containerized Solar-Hybrid Wellhead Gas-Lift Optimizer

The ND-GLO (Niger Delta Gas-Lift Optimizer) is a containerized, solar-hybrid surface system that transforms flared or stranded gas into productive artificial lift energy. The system is designed for autonomous operation in remote marginal fields where grid power and SCADA infrastructure are unavailable.

Key differentiators include: (1) gas-lift-first conditioning that prioritizes lift gas over flare, (2) auto-bypass logic that maintains production during compressor maintenance, (3) H2S cartridge treatment that protects downstream equipment without complex chemical injection systems, and (4) solar-hybrid power that eliminates diesel generator dependency. The entire system fits within a standard 20-foot shipping container for rapid deployment and relocation.

Key Claims:

- Integrated containerized gas-lift system
- Gas-lift-first conditioning with auto-bypass logic
- Solar-hybrid power architecture with battery backup
- H2S cartridge treatment module
- Autonomous operation without SCADA dependency
- Containerized portability and rapid deployment

Patent Status: Provisional filed June 13, 2026. Multiple claims. PCT strategy document prepared with WIPO fee reduction pathway.

Integration: Surface complement to Niger Delta Survivor. Together they form a complete downhole-to-surface production optimization platform.

03. ND-RIP

ND-Reservoir Intelligence Probe with Proprietary Edge AI Processor

The ND-RIP is a next-generation downhole diagnostic probe that brings laboratory-grade reservoir characterization to the wellbore in real time. It integrates multiple specialized sensor modules — including resistivity arrays, nuclear magnetic resonance, pulsed neutron gamma, distributed strain monitoring, quantum pressure-temperature, ultrasonic fluid imaging, and hydrogen gas detection — all orchestrated by a proprietary edge AI processor.

The proprietary edge AI processor features a high-performance application core, a dedicated neural processing unit for NDIL-PINN v3.1 inference, an FPGA for real-time signal processing, and a complete memory subsystem with ECC protection. Power management supports geothermal harvesting and battery backup. The system runs the ND-RIP Physics-Informed RL Engine for adaptive decision-making under uncertainty.

Key Claims:

• Multi-module integrated sensor architecture

• Proprietary edge AI processor with dedicated NPU and FPGA

• Real-time NDIL-PINN v3.1 edge inference

• Physics-Informed RL Engine for autonomous control

• Geothermal power harvesting plus battery backup

• Standard model export pipeline for edge deployment

Patent Status: Provisional filed June 13, 2026. Non-convention Nigerian patent application in progress.

04. ND-AMAHOR

AI-Native Artificial Lift Intelligence Platform

ND-Amahor is a unified, AI-native artificial lift platform designed to compete directly with industry-standard tools including PIPESIM, PROSPER, RODSTAR, and XROD. Unlike legacy software that relies on steady-state physics and manual parameter tuning, ND-Amahor uses Physics-Informed Neural Networks (PINNs), Reinforcement Learning control agents, and multi-agent swarm coordination to deliver autonomous, real-time lift optimization.

The platform architecture comprises 5 layers: (1) Meta-PINN with adaptive physics weighting, (2) Soft Actor-Critic (SAC) RL control agent, (3) Multi-Agent RL swarm coordination, (4) Blockchain audit trail via Hyperledger, and (5) 3D digital twin visualization. The system covers multiple integrated physics domains and is deployable on edge devices for wellsite operation.

Key Claims:

- Meta-PINN adaptive physics weighting across multiple domains
- SAC reinforcement learning control for ESP and gas lift
- Multi-Agent RL swarm coordination
- Hyperledger blockchain audit trail
- 3D digital twin visualization
- Edge deployment on industry-standard hardware
- Modular architecture with extensive API coverage

Funding: Seeking growth capital to complete development and conduct field trials. 12-month development roadmap.

Competitive Position: First AI-native integrated lift platform with physics-informed RL and blockchain audit for the African market.

05. 5A AUTONOMOUS LOOP

Acquire - Analyze - Assimilate - Anticipate - Act

The 5A Autonomous Loop is the holistic control architecture that orchestrates the entire ND innovation platform. It provides a unified framework for autonomous decision-making under uncertainty, closing the loop between reservoir sensing, physics-informed modeling, predictive analytics, and autonomous well intervention.

Acquire: Real-time data ingestion from ND-RIP sensors, surface SCADA, and third-party sources. **Analyze:** NDIL-PINN v3.1 physics-informed inference and anomaly detection on edge hardware. **Assimilate:** Bayesian neural network world model updates with new field data and synthetic realizations. **Anticipate:** Predictive forecasting of production decline, equipment failure, and reservoir sweep efficiency. **Act:** Safe action filtering through the SAC RL agent with physics-informed reward shaping and regulatory constraints.

Key Claims:

• Holistic 5-layer autonomous control architecture

• Physics-informed reward shaping for safe RL

• Adaptive entropy tuning for exploration-exploitation balance

• Safe action filter with regulatory constraint enforcement

• Closed-loop integration with ND-RIP, ND-Amahor, and ND-GLO

Patent Status: Provisional filed June 13, 2026. Covers the full autonomous loop system including proprietary edge AI processor deployment.

COMPETITIVE LANDSCAPE

The ND platform occupies a unique position in the African O&G technology market. No existing competitor offers an integrated stack that combines downhole hardware, surface gas-lift optimization, AI-native lift modeling, and autonomous reservoir intelligence within a single indigenous framework.

Capability	ND Platform	PIPESIM/PROSPER	RODSTAR/XROD	Generic SCADA
Downhole Hardware	Yes (Survivor)	No	No	No
Surface Gas-Lift	Yes (ND-GLO)	Model only	N/A	Monitor only
AI-Native Modeling	Yes (PINN + RL)	No	No	No
Autonomous Control	Yes (5A Loop)	No	No	Alarm only
Edge Deployment	Yes (Proprietary)	No	No	Partial
Blockchain Audit	Yes (Hyperledger)	No	No	No
Indigenous Design	Yes (Nigeria)	No	No	Varies

FINANCIAL PROJECTIONS & FUNDING

The ND platform is structured for a growth capital funding round to complete development, conduct field trials, and achieve commercial readiness within 24 months. The following projections are based on conservative assumptions for the West African marginal field market.

Phase	Timeline	Funding	Milestone
Development	Q3 2026 - Q2 2027	Growth Capital	Complete ND-Amahor v1.0, field trial 3 wells
Scale	2028-2029	TBD	Regional deployment, 50+ well installations
Revenue Model	2027+	N/A	Licensing (industry-standard terms) + SaaS subscription (ND-Amahor)
Target TAM	West Africa	2.8B USD	Marginal field production optimization market

PARTNERSHIP & LICENSING OPPORTUNITIES

The ND platform is available for collaboration under structured partnership and licensing agreements. All engagements begin with a signed NDA followed by a Letter of Intent defining exclusivity, royalty terms, and minimum commercialization commitments.

Fabrication Partnership (Niger Delta Survivor):

Exclusive license for indigenous manufacturing under industry-standard licensing terms. Targeting leading indigenous fabricators. Partial technical specifications released under NDA; advanced patentable features withheld until full patent filing is complete.

Field Trial Partnership (ND-GLO):

Seeking marginal field operators for pilot deployment. Partner provides well access and operational support; ND team provides equipment, installation, and monitoring. Revenue share model post-validation.

Investment (ND-Amahor / ND-RIP):

Growth capital via equity or convertible note structure. Investor receives board observer seat, quarterly technical reviews, and first right of refusal on future rounds. Use of funds: 40% R&D, 30% field trials, 20% IP/legal, 10% operations.

Employment (Australia 482 TSS):

Seeking 482 TSS employer sponsorship for FIFO drilling or production operations in Western Australia or Queensland. Full employer sponsorship required including visa nomination and relocation support. Prepared to commit to minimum employment term. Upstream oil & gas roles only — not mining.

REGULATORY COMPLIANCE & RISK MITIGATION

The ND platform is designed for full compliance with Nigerian upstream regulatory frameworks including NUPRC (Nigerian Upstream Petroleum Regulatory Commission), NDPR (Nigeria Data Protection Regulation), PIA (Petroleum Industry Act), and NCDMB (Nigerian Content Development and Monitoring Board) local content requirements.

IP Protection:

• Multiple provisional patents filed (priority date: June 13, 2026)

• 12-month priority window for full non-convention and PCT filings

• NDA/LOI templates prepared under Nigerian law with LCA arbitration

• Substantial liquidated damages clause for confidentiality breach

• All external collaboration preceded by signed NDA

Technical Risk Mitigation:

• Synthetic data generation plan addresses scarce historical well data for PINN training

• Extensive stochastic dataset with validation against published Niger Delta field data

• Comprehensive implementation roadmap for complete training dataset assembly

• Modular architecture allows component-level validation before system integration

• Field trial program designed with staged gate reviews and kill-switch criteria

CONTACT & NEXT STEPS

I welcome inquiries from investors, fabrication partners, field operators, SPE collaborators, and Australian employers. All initial discussions are protected under mutual NDA.

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SPE Connect	Active member - innovation posts and technical discussions

Recommended Next Steps for Interested Parties:

- Investors: Request the investor Q&A; document and schedule a technical demo of NDIL-PINN v3.1
- Fabrication Partners: Execute NDA and receive partial fabrication specifications
- Field Operators: Schedule wellsite assessment for ND-GLO pilot suitability evaluation
- Australian Employers: Review CV and confirm ANZSCO alignment for 482 TSS nomination
- SPE Collaborators: Connect on SPE Connect or request the ND-Survivor presentation